PACKAGE INSERT

ZORAX SUSPENSION

Each 5ml contains:
Acyclovir 200mg

Product Description: Creamy, white suspension with banana flavour.

Pharmacodynamics:
Acyclovir is a synthetic acyclic purine nucleoside analogue with in vitro inhibitory activity against Herpes simplex type 1 and 2 (HSV-1 and HSV-2), Varicella-zoster, Epstein-Barr and cytomegalovirus. In cell cultures, the inhibitory activity of acyclovir for Herpes simplex virus is highly selective. Cellular thymidine kinase does not effectively utilize acyclovir as a substrate. Herpes simplex virus-coded thymidine kinase, however, converts acyclovir into acyclovir monophosphate, a nucleotide. The monophosphate is further converted into diphosphate by cellular guanylate kinase and into triphosphate by a number of cellular enzymes. Acyclovir triphosphate interferes with Herpes simplex virus DNA polymerase and inhibits viral DNA replication. Acyclovir triphosphate also inhibits cellular α-DNA polymerase but to a lesser degree. In vitro, acyclovir triphosphate can be incorporated into growing chains of DNA by viral DNA polymerase and to a much smaller extent by cellular α-DNA polymerase. When incorporation occurs, the DNA chain is terminated. Acyclovir is preferentially taken up and selectively converted to the active triphosphate form by herpes virus infected cells. Thus, acyclovir is much less toxic in vitro for normal uninfected cells because: 1) less is taken up; 2) less is converted to the active form; 3) cellular α-DNA polymerase is less sensitive to the effects of the active form.
Herpes simplex virus develops resistance to acyclovir in vitro and in vivo due to the selection of mutants deficient in thymidine kinase. Other mechanisms of resistance include altered specificity of thymidine kinase and reduced sensitivity of viral DNA polymerase. Resistant viruses are most likely to be a problem in patients with a suppressed immune response. AIDS patients may be particularly prone to acyclovir-resistant mucocutaneous herpes simplex virus infections.

Pharmacokinetics:
About 15 to 30% of a dose of acyclovir given by mouth is considered to be absorbed from the gastro-intestinal tract. A dose of 200mg acyclovir every 4 hours by mouth is reported to produce maximum and minimum steady-state plasma concentrations of 0.56 and 0.29mcg per ml respectively; equivalent values following 400mg doses are 1.2 and 0.6mcg per ml.
Acyclovir is excreted through the kidney by both glomerular filtration and tubular secretion. The terminal or beta-phase half-life is reported to be about 2 to 3 hours for adults without renal impairment. In chronic renal failure this value is increased and may be up to 19.5 hours in anuric patients. During haemodialysis the half-life is reduced to 5.7 hours, with 60% of a dose of acyclovir being removed in 6 hours. In the elderly, total body clearance falls with increasing age associated with decreases in creatinine clearance although there is little change in the terminal plasma half-life. Most of a dose by intravenous infusion is excreted unchanged with only up to 14% appearing in the urine as the inactive metabolite 9-carboxymethoxymethylguanine. Faecal excretion may account for about 2% of a dose. There is wide distribution to various tissues, including the CSF where concentrations achieved are about 50% of those achieved in plasma. Protein binding is reported to range from 9 to 33%. Prior administration of probenecid increases the half-life and the area under the plasma concentration-time curve of acyclovir.

Indication:
Acyclovir is indicated for:-
a) Treatment of Herpes simplex virus infections of the skin and mucous membrane, including initial and recurrent herpes genitalis and herpes labialis.
b) Suppression of recurrent Herpes simplex infections in immune-competent patients.
c) Prophylaxis of Herpes simplex infections in immune-compromised patients.
d) Treatment of varicella (chicken pox) and Herpes zoster (Shingles) infection.

Recommended Dosage:
<u>Treatment of Herpes Simplex Infections</u>
The usual dose by mouth is 200mg(5ml) five times daily (usually every 4 hours while awake) for 5 to 10 days.
In the severely immunocompromised patient, the dose is 400mg (10ml) four times daily (every 6 hours).
Dosing should begin as early as possible after the start of an infection; for recurrent episodes this should preferably be during the prodromal period or when lesions first appears.

<u>Suppression of Herpes Simplex Infections</u>
The dose by mouth is 200mg (5ml) two to five times daily, interrupted every 6 to 12 months for reassessment of the condition; alternatively 400mg (10ml) may be given twice daily.
Chronic suppressive treatment is not suitable for mild or infrequent recurrences of herpes simplex infection. In such case, episodic treatment of recurrences may be more beneficial.

<u>Prophylaxis of Herpes Simplex Infections</u>
For prophylaxis of Herpes simplex infections in immune-compromised patients, 200mg (5ml) acyclovir should be taken four times daily at approximately six-hourly intervals.
In severely immune-compromised patients (e.g. after marrow transplant) or in patients with impaired absorption from the gut the dose can be doubled to 400mg (10ml) acyclovir four times daily at approximately six hourly intervals or, alternatively, intravenous dosing could be considered. The duration of prophylactic administration is determined by the duration of the period at risk.

<u>Treatment of Varicella and Herpes Zoster Infections</u>
The usual dose for treatment of zoster infections is 800mg (20ml) of acyclovir five times daily (every 4 hours while awake) for 7 to 10 days. In severely immune-compromised patients (e.g. after marrow transplant) or in patient with impaired absorption from the gut, consideration should be given to intravenous dosing.
Dosing should begin as early as possible after the start of an infection: treatment yields better results if initiated as soon as possible after onset of the rash.

<u>Dosage in Children</u>
For treatment of Herpes simplex infections, and for prophylaxis of Herpes simplex infections in the immune-compromised, children aged two years and over should be given adult dosages and children below the age of two years should be given half the adult dose.
For treatment of Varicella infections, children over the age of six years can be given 800mg (20ml) acyclovir four times daily and children between the ages of two and six years can be given 400mg (10ml) acyclovir four times daily. Children below the age of two years can be given 200mg (5ml) acyclovir four times daily. Dosing may be more accurately calculated as 20mg acyclovir /kg bodyweight [not to exceed 800mg (20ml)] four times daily. Treatment should continue for five days. No specific data are available on the suppression of Herpes simplex infections or the treatment of Herpes zoster infections in immune competent children.

<u>Dosage in the Elderly</u>
In the elderly, total acyclovir body clearance declines in parallel with creatinine clearance. Adequate hydration of elderly patients taking high oral doses of acyclovir should be maintained. Special attention should be given to dosage reduction in elderly patients with impaired renal function.

<u>Dosage in Renal Impairment</u>
In the management of herpes simplex infections in patients with impaired renal function, the recommended oral doses will not lead to accumulation of acyclovir above levels that have been established safe by intravenous infusion. However, for patients with severe renal impairment (creatinine clearance less than 10ml/minute) an adjustment of dosage to 200mg (5ml) twice daily at approximately twelve-hourly intervals is recommended.
In the treatment of Varicella and Herpes zoster infections it is recommended to adjust the dosage to 800mg (20ml) twice daily, at approximately twelve-hourly intervals, for patients with severe renal impairment (creatinine clearance less than 10ml/minute) and to 800mg (20ml) three times daily, at intervals of approximately eight hours, for patients with moderate renal impairment (creatinine clearance in the range 10 to 25ml/minute).

Route of Administration: Oral

Contraindications:
It is contraindicated in patients with known hypersensitivity to acyclovir and propylene glycol.

Warnings and Precautions:
All patients should be cautioned to ensure they avoid the potential of virus transmission, particularly when active lesions are present while they are on therapy. In severely immune-compromised patients the physician should be aware that prolonged or repeated courses of acyclovir may result in selection of resistant viruses which may not fully respond to continued acyclovir therapy.

Interactions with Other Medicaments:
Co-administration of probenecid with intravenous acyclovir has been shown to increase the mean half-life and the area under the concentration-time curve. Urinary excretion and renal clearance were correspondingly reduced. The clinical effects of this combination have not been studied.
Other drugs affecting renal physiology could potentially influence the pharmacokinetics of acyclovir. However, clinical experience has not identified other drug interactions with acyclovir.

Pregnancy and Lactation:
<u>Teratogenicity & Pregnancy</u>
Acyclovir was not teratogenic in standard tests in mouse, rabbits or rats. In a non-standard test in rats, there were fetal abnormalities and maternal toxicity. There are no adequate and well-controlled studies in pregnant women. Acyclovir should not be used during pregnancy unless the potential benefit justifies the potential risk to the fetus. Although acyclovir was not teratogenic in standard animal studies, the drug's potential for causing chromosome breaks at high concentration should be taken into consideration in making this determination.

<u>Lactation</u>
Acyclovir concentrations have been documented in breast milk in two women following oral administration of acyclovir and range from 0.6 to 4.1 times corresponding plasma levels. These concentrations would potentially expose the nursing infant to a dose of acyclovir up to 0.3mg/kg/day. Caution should be exercised when acyclovir is administered to a nursing woman.

Side Effects:
Acyclovir given by mouth may cause gastro-intestinal effects such as nausea, vomiting, and diarrhoea; headache and skin rashes have also been reported.
Reversible neurological reactions, notably dizziness, confusional states, hallucinations and somnolence, have occasionally been reported, usually in patients with renal impairment or other predisposing factors. Other events reported rarely in patients include mild, transient rises in bilirubin and liver-related enzymes, small increases in blood urea and creatinine, small decreases in haematological indices, headache and fatigue.

Symptoms and Treatment of Overdose:
Acyclovir is only partially absorbed in the gastrointestinal tract. No acute massive overdosage of orally administered acyclovir has been reported. Doses as high as 800mg (20ml) 6 times daily for 5 days have been administered to humans without acute untoward effects.
Treatment: Ingestion of doses of acyclovir in excess of 5g warrants close observation of the patients. Acyclovir is dialysable by haemodialysis.

Storage Conditions: Store at or below 30°C. Protect from light.

Pack Sizes: A bottle of 60ml, 100ml, 120ml, 200ml and 250ml.
Pack Sizes (export only): A bottle of 1 litre.

Shelf-life: 1.5 years.

FURTHER INFORMATION CONCERNING THIS DRUG CAN BE OBTAINED FROM YOUR FAMILY PHYSICIAN / LOCAL GENERAL PRACTITIONER / PHARMACIST.

Manufacturer & Product Registration Holder:
SUNWARD PHARMACEUTICAL SDN. BHD.
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